

Quiz (Carolina Reaper)

- Q1.** If $f(x) = x^2$ and $g(x) = 2x - 1$, what is $f(g(x))$?
(A) $4x^2 - 4x + 1$
(B) $4x^2 - 2$
(C) $2x^2 - x$
(D) $x^2 + 2x - 1$
(E) $4x^2 + 1$
- Q2.** A cargo ship travels from port A to port B at a speed of 20 km/h. If the distance between the ports is d km, and $f(x) = 2x$ and $g(x) = x - 5$, what is the domain of $f(g(d))$ if the ship cannot travel more than 35 km?
(A) $[5, 20]$
(B) $[0, 15]$
(C) $[5, 35]$
(D) $[0, 30]$
(E) $[5, 25]$
- Q3.** A company uses the function $f(x) = 3x + 2$ to calculate the cost of shipping x packages, and $g(x) = x^2$ to calculate the number of packages that can be shipped in x days. What is $f(g(2))$?
(A) 16
(B) 20
(C) 14
(D) 12
(E) 18
- Q4.** The traffic flow on a highway can be modeled using the function $f(x) = 2x + 1$, where x is the number of cars on the road. If $g(x) = x - 3$ represents the number of cars that leave the road, what is $f(g(5))$?
(A) 11
(B) 9
(C) 7
(D) 5
(E) 3
- Q5.** A delivery truck uses the function $f(x) = x^2 + 1$ to calculate the time it takes to travel x miles. If $g(x) = 3x - 2$ represents the number of miles the truck travels in x hours, what is $f(g(1))$?
(A) 10
(B) 8
(C) 12
(D) 6
(E) 4
- Q6.** If $f(x) = x^2 - 2$ and $g(x) = 2x + 1$, what is the domain of $f(g(x))$ if x cannot be greater than 3?
(A) $[-1, 3]$
(B) $[-2, 2]$
(C) $[-1, 2]$
(D) $[-2, 3]$
(E) $[-3, 1]$
- Q7.** A shipping company uses the function $f(x) = 2x + 3$ to calculate the cost of shipping x packages. If $g(x) = x^2 - 1$ represents the number of packages that can be shipped in x days, what is $f(g(2))$?
(A) 13
(B) 11
(C) 9
(D) 15
(E) 7
- Q8.** The function $f(x) = x^2 + 2$ models the time it takes for a package to be delivered x miles. If $g(x) = 2x - 3$ represents the number of miles the package travels in x hours, what is $f(g(2))$?
(A) 15
(B) 13
(C) 11
(D) 9
(E) 17

Open Ended Questions (FRQ)

- Q9.** A transportation company uses the function $f(x) = 3x + 1$ to calculate the cost of shipping x tons of cargo. If $g(x) = 2x^2 - 5$ represents the number of tons of cargo that can be shipped in x days, find $f(g(x))$ and determine the domain restrictions if the company can only ship cargo for 5 days.
- Q10.** The traffic flow on a highway can be modeled using the function $f(x) = 2x + 1$, where x is the number of cars on the road. If $g(x) = x - 2$ represents the number of cars that leave the road, find $f(g(x))$ and interpret the result in the context of traffic flow. Determine the domain restrictions if the road can only accommodate 10 cars.

Chef's Tips

- Use a graphing calculator to visualize composite functions
- Check for domain restrictions when evaluating composite functions
- Use real-world contexts to interpret composite functions